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Design and performance assessment of custom static intermixers in extrusion 3D printing using machine learning–driven image analysis

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Uniform blending in multi-material extrusion additive manufacturing is crucial for ensuring consistent material properties. This study evaluates the performance of five static intermixer designs, Split Path, Helix Array, Full Turn Helix, Half Moon, and Cross Bars, integrated into a coaxial extruder system for enhancing the blending of multi-colored polylactic acid (PLA) pellets. Each mixer was tested using a 50/50 mixture of red and blue PLA under controlled extrusion conditions at 210 °C. Mixing performance was assessed through microscopic imaging and machine learning-based analysis, including histogram evaluation, clustering algorithms, and standard color uniformity indices. Results showed that the Split Path and Full Turn Helix mixers provided the most uniform color distribution, with minimal segregation. In contrast, the Helix Array, Half Moon, and Cross Bars designs produced moderate to inconsistent mixing, showing visible streaking and uneven blending. All mixer configurations, however, significantly outperformed the control (no mixer) setup. These findings offer quantitative insights into the effectiveness of various mixer geometries, providing a basis for optimizing mixing strategies in multi-material extrusion additive manufacturing. The study contributes to the development of more reliable extrusion systems for applications such as functionally graded materials, and advanced polymer composites.

Keywords Pellet extrusion, Additive manufacturing, Mixer, Split path, Mahor extruder, Blending, Machine learning, Histogram analysis, Cluster analysis, Standard index analysis

Additive manufacturing (AM), also known as 3D printing, has the ability to produce complex, custom-designed structures with high precision which helped in revolutionizing modern manufacturing techniques. AM technologies are set in seven categories including Extrusion and Vat Photo-polymerization[1], among which multi-material extrusion (MME) stands out as a strong method for combining different materials into a single printed part, providing customized mechanical, thermal, and aesthetic properties[2]. That specific benefit is particularly beneficial in soft robotic, biomedical implant, flexible electronic device, and polymer composite applications, in which functionally graded materials (FGMs) and continuous gradation between disparate materials become critical for performance maximization and durability improvement[3]. Nevertheless, even with its significant potential, MME faces significant challenges in terms of uniform mixing of materials, stemming from discrepancies in viscosities, flow behavior, and thermal conductivities of materials; therefore, such complications can cause defects, such as phase segregation, poor interfacial adhesion, and variation in structural integrity[1].

A key challenge in multi-material printing is regulating material flow behavior, with polymers with disparate viscosities tending to exhibit unbalanced extrusion dynamics[4,5]. For example, a high viscosity polymer such as acrylonitrile butadiene styrene (ABS) will not combine with a low-viscosity elastomer such as thermoplastic polyurethane (TPU) in an acceptable manner unless careful extrusion settings are followed[6]. In a similar scenario, viscosity mismatches can result in poor adhesion at interfaces and delamination[7], specifically in structures that

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require integrity, such as impact-bearing biomedical implants and impact-bearing automotive parts[8]. In addition, phase segregation can occur when immiscible materials, such as hydrophobic and hydrophilic polymers, cannot homogeneously mix, producing defects such as air bubbles and poor mechanical performance[9]. Overcoming such obstacles is significant in terms of delivering functional integrity for multi-material printed parts.

The research is in progress to enhance the blending efficiency. A variety of mixing mechanisms have been introduced and examined, such as optimized nozzles, static mixers and dynamic mixers[10,11]. for instance, the Split-path and helical static mixers were found to produce shear forces due to their internal structures, which allows for homogenization of materials. Helical mixers have proved their ability to produce turbulent flow regimes, which enhances blending capacities, and allows for proper mixing of materials that have similar viscosities. Split-path mixers, in contrast, use a variety of flow routes for materials, producing reduced phase segregation and increased uniformity[12,13]. On the other hand, dynamic mixers use moving parts to produce additional shear forces, allowing for efficient blending of materials with high viscosity differences[14]. Extrusion nozzle configuration is also important in blending efficiency; nozzle structures such as converging–diverging nozzles can effectively manage flow regimes for increased mixing performance.

In addition to mixer design, extrusion parameters such as temperature, feed rate, and pressure have a significant impact on mixing efficiency[15]. Knowing that higher temperatures during extrusion results in decreasing the viscosity of the material, which in return enhances the flow and adhesion between layer interfaces. However, extreme temperatures result in material degradation which causes non-uniform mixing and poor mechanical properties[15,16]. For instance, high-performance materials such as polyether ether ketone (PEEK) demand careful temperature control in order to avert thermal degradation with uniform distribution of materials. Likewise, fluctuations in feed rate can produce fluctuations in material flow, creating non-uniform distribution and development of voids. Tuning these settings is critical in order to produce uniform mixing and preserve desired multi-material print properties[17].

Assessing the homogeneity and distribution of materials entails powerful analysis techniques providing quantitative information regarding distribution and homogeneity of materials. With the aid of microscope visualization, coupled with computing algorithms for image processing, researchers can observe the mixing on a high resolution and are able to spot differences on a micro scale[18]. These differences can then be quantitatively examined using such algorithms as the Structural Similarity Index (SSIM) and Mutual Information (MI) which are used to evaluate the colors' distribution and homogeneity of materials within the extruded parts[19]. Image based numerical modeling like the finite element analysis (FEA) allows for the simulation of flow behavior and shear rates in intricate mixer designs as well as allows researchers to optimize extrusion systems[20].

The uniform distribution of components in MME is critical for modern materials as well as parts for employment in different industries. For instance, in functionally graded materials (FGMs), mixing is done to achieve a very gradual transition from one material to another, which often have extreme differences, like the soft and hard regions in biomedical implants or the heat resistant and low-weight sections in aerospace parts[21]. Moreover, blending must be of a precision level in wires and circuits for wearable devices, where stretchable sensors are used to join conductive and insulation materials for electronic and mechanical performance multifunctional devices[22]. This is true for polymeric composites whereby blending materials is done to increase not only strength, but other mechanical properties such as impact strength, and toughness, in order the widen the application area to automotive engineering and performance structures.

The developments that are now transforming the course of additive manufacturing will facilitate the making of highly functional, custom-made, high-performance components of the different industries. Engineers and researchers can enhance the practicality of multi-material 3-D printing by resolving blending issues via advanced mixer designs, innovative control techniques, and evaluation methods[23]. With the continuous evolution of the AM field, there is a significant need for further research in optimized blending mechanisms and in real-time process monitoring to act as a major driving force for the innovation of multi-material fabrication.

In this research, quantitatively analyzed the uniformity of blending, and machine learning techniques were applied, including histogram analysis, clustering methods, and standard index evaluations. These computational approaches enabled an objective assessment of color distribution and material homogeneity across different extruded specimens. This study highlights the critical role of structured mixing mechanisms in improving material homogeneity during multi-material extrusion. The insights gained from this research provide a foundation for optimizing mixer designs to enhance blending efficiency, which is crucial for applications in functional polymer processing, gradient materials, and advanced manufacturing systems. The findings contribute to the broader goal of improving multi-material extrusion techniques for high-performance additive manufacturing applications.

Materials and methodology

Materials

The materials used in the project vary from the manufacturing and prototyping stage to the testing stage. The intermixer parts were printed using a High Temperature resistance resin (EPAX 3D, USA). The mixers were printed using an LCD resin printer (Make: ELEGOO, Model: Mars 3 Pro, https://www.elegoo.com/en-ca, Shenzhen, China). The hot end block was crafted from an Aluminum (Al) block. The material fed into the Mahor extruder was Polylactic acid (PLA) in two different colors: red and blue.

Extruder design

The pellet extrusion head to be used is a Mahor V4 extruder, as shown in Fig. 1a. Plastic pellets are fed into the print head via a hopper, and as they move through the heat block, they melt before being extruded through the nozzle. The extruder is ideal for testing various geometries due to its removable nozzle, allowing the attachment of an extended hot end capable of accommodating different mixing geometries. Moreover, it supports a range of

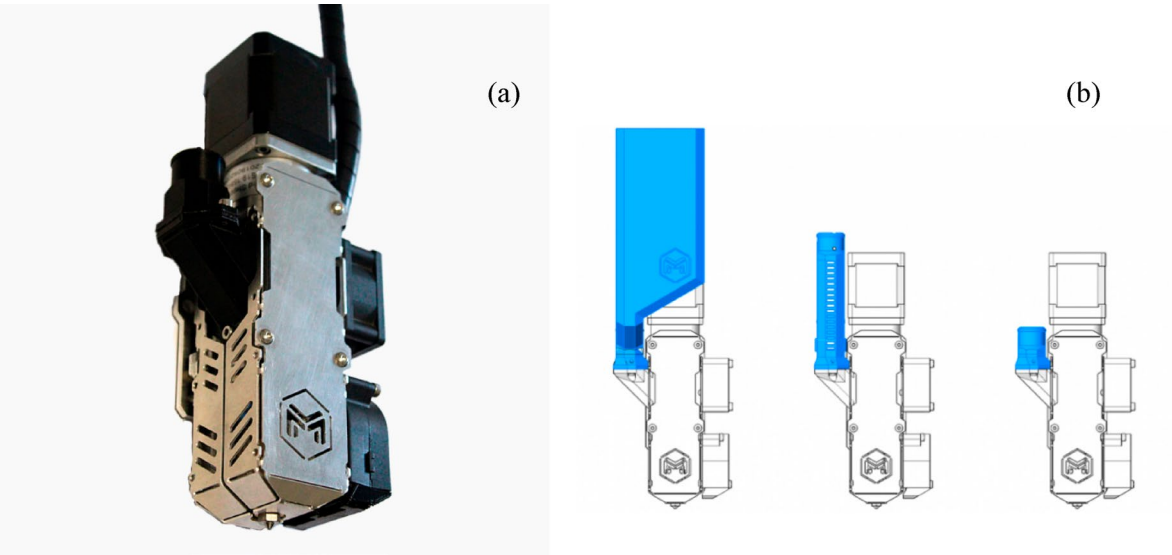


Fig. 1. (**a**) Mahor V4 Extrusion Head, and (**b**) different Pellet Input Methods.

pellet input methods, as seen in Fig. 1b. The custom hot end must be compatible with the output of the Mahor V4 extruder and fit within its aluminum shielding, which measures 36 mm × 36 mm. While a shorter print head is preferred to minimize the reduction in maximum print height, the primary goal is to prove the concept of static mixing rather than creating an optimized production unit. The components required within the geometry containment unit include mixing geometries, a cylindrical cartridge heater, and any additional parts necessary for proper functionality. The selected design features two halves, a top, and a bottom, clamped over the mixing geometry to secure it in place

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Custom hot-end

The hot-end design is shown in Fig. S1. The mixing geometries have a cylindrical shape, 15 mm in diameter and 40 mm in length. Given this size, the top and bottom halves of the containment unit should include a cavity large enough to securely fit the geometries. Additionally, 5 mm of material will be allocated above and below the geometry cavity for threading to accommodate the input and output connections of the device. The heater hole was designed to pass completely through the assembly, from the top of the upper block to the bottom of the lower block, allowing the heater to be installed from either direction. A shoulder was added to the threaded input hole in the top block to provide a seating surface for the threaded transfer tube. This design minimizes the risk of leaks and prevents over-threading, which could potentially cause the tube to contact and damage the geometry being tested. Assembly views of the final mixer hot end are shown in Fig. S1.

Intermixer design

Intermixer geometries were created using Solid Edge CAD. These shapes were strategically arranged, patterned, and mirrored in the assembly environment to form various intermixer designs. Each mixer was designed to fit the containment unit's internal chamber, with dimensions of approximately 40 mm in height and 15 mm in diameter. Several initial design concepts were generated based on collective intuition, with only the most feasible configurations being selected for further exploration. Five final design alternatives were selected for experimentation, as shown in Fig. 2a–e.

The aim is to identify the optimal mixer for effectively mixing red and blue PLA, using machine learning-based image processing techniques applied to images of the mixed material surfaces. The effectiveness of blending is analyzed by examining cross-sections of the samples through microscopic imaging and image processing techniques. Based on both quantitative and qualitative results, the best-performing intermixer designs are identified for further development in future research. The analysis involves three major categories of techniques to evaluate and compare mixer performance comprehensively. Histogram analysis techniques[24] are used to analyze the color distribution within the images through histograms, employing various indices to measure the position, overlap, and representation of the histogram data. This step quantifies how well the colors are distributed and mixed, providing a visual and numerical understanding of the mixing process. Clustering techniques[25] are applied to group image color pixels into distinct clusters, representing different levels of mixing, and multiple clustering methods are utilized to determine the optimal number of clusters, reflecting the complexity and uniformity of the color distribution. By analyzing the clustering results, the degree of color blending and the effectiveness of the mixers are evaluated. Finally, standard measuring indexes[26] are calculated to assess the similarity and deviation between pixel intensities in the images, with metrics such as pixel similarity, standard deviation, and mean intensity used to quantify the homogeneity of the color mixture.

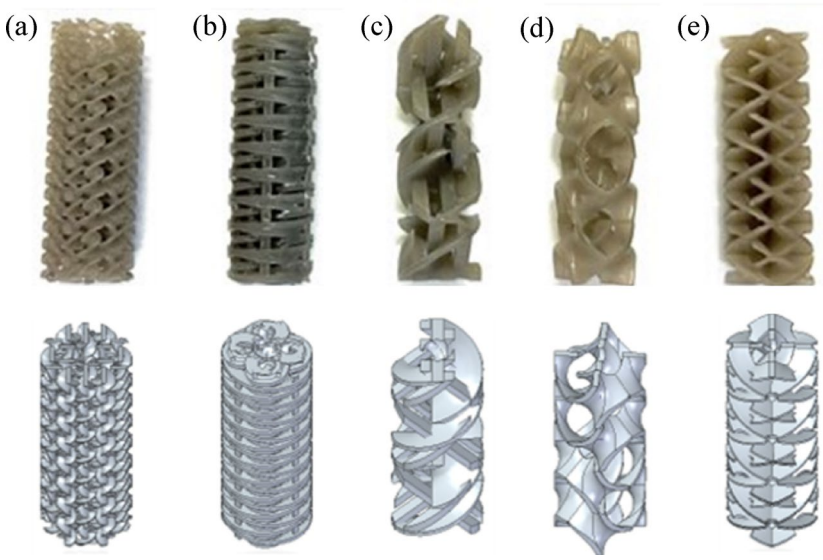


Fig. 2. Intermixer designs, (**a**) Helix Array design, (**b**) Split Path Mixer design, (**c**) Cross Bars design, (**d**) Half Moon design, and (**e**) Full Turn Helix.

These methodologies collectively provide a robust framework for evaluating and identifying the best-performing mixer for mixing red and blue PLA, using advanced image processing and analysis techniques.

Helix array intermixer design
Figure 2a shows the Helix Array Intermixer, which is made up of several layers each with the base unit of tightly packed 180° spiral shapes (helices) stacked next to each other. The design of each layer is based on the pattern seen in cross-sections of wire ropes. Each new layer is rotated 90 degrees from the one below it. This creates a multi-scale arrangement that helps divide the flow as much as possible. Since thermoplastics behave in a non-Newtonian way, they don't mix well through turbulence or swirling motions. Instead, this design focuses on repeatedly splitting and rejoining the flow to blend the materials effectively.

Split path intermixer design
The Split Path Intermixer, shown in Fig. 2b, was originally built around a central helical (spiral) structure. Helical shapes are commonly used in mixing because they help push material outward, improve mixing consistency, maintain smooth flow paths, and work well with 3D printing. This design includes four main helices, each with a smaller helix inside that varies in pitch and direction of flow. The main helices are rotated 90 degrees from each other along the vertical (z) axis, allowing their shapes to interlock at different points. To prevent slow-moving areas, triangular columns were added in the larger gaps between the outer helices and the mixer wall. A spiral-shaped outer shell also surrounds the entire mixer to help guide the material toward the center. Altogether, this design uses complex helical paths to create many flow routes and improve how evenly materials are mixed.

Cross bars intermixer design
The Cross Bars Intermixer, shown in Fig. 2c, is based on mixing designs commonly used in industry. It consists of angled crossbars placed at 45°, which block and redirect the flow, forcing the material to split and rejoin at each point. To handle the expansion of thermoplastic as it leaves the extruder and enters the mixing chamber, helical patterns offset by 90° were added between each set of crossbars. These features help spread the material more evenly across the chamber, increase how far the fluid travels, and create more opportunities for the flow to be divided horizontally, which ultimately improving the overall mixing.

Half-moon intermixer design
Figure 2d shows the Half-Moon Intermixer, which uses multiple helical elements to first pull the flow inward toward the center, then push it back outward. A central helical pattern, rotated 90° at each level, helps split the flow evenly across the mixer. One of the main goals of this design is to increase the internal space (cavity volume) so that thermoplastic can spread out more under lower pressure. Even though the pressure is reduced, the added helical paths extend the flow distance, helping to ensure the material mixes thoroughly.

Full turn helix intermixer design
The Full Turn Helix Intermixer, shown in Fig. 2e, uses multiple layers of helical patterns that are rotated 90° and span the entire width of the chamber. A smaller set of offset helices is placed in the center to help guide the flow through the core. To prevent the material from slowing down near the chamber walls, small circular holes were added where the outer helices intersect. The main goal of this design is to create unpredictable flow paths and

multiple points where the material splits and rejoins—both near the edges and at the center. This helps increase horizontal mixing and gives the thermoplastic more time to interact and blend inside the chamber.

Intermixer fabrication

Key design iterations focused on the following improvements: a minimum wall thickness of 0.75 mm to prevent mixer component fractures, and a clearance of approximately 1 mm to avoid flow blockages. Potential stagnation points were eliminated by adding cutouts for flow bypass. Various mixer geometries were created to assess which parameters optimize blending. Additionally, spherical cutouts (3–10 mm in diameter) were added at the mixer's entry and exit points to reduce pressure buildup by preventing nozzle blockages. The printed prototypes can be seen in Fig. 2.

The five designs were printed using EPAX high-temperature resin, which can withstand temperatures up to 220 °C (EPAX 3D, 2023). The printing process was carried out with an LCD resin printer (Make: ELEGOO, Model: Mars 3 Pro, Shenzhen, China). Each of the final five design alternatives is printed using high-temperature resin and subjected to experimental testing. Performance data and extruded samples are collected for each intermixer design.

Image capturing

After producing the samples, specimens were cut and analyzed for short-range blending consistency using microscopic imaging. The specimens, ranging from 5 to 10 mm in length, were examined at both cross-sectional ends. A total of 90 specimens were collected and imaged with an OMAX 4X-100X microscope, along with a USB digital camera, to capture detailed cross-sections.

Experimental setup

System assembly was done in a way that a Mahor V4 Pellet Extruder was used to melt PLA pellets and push them through the mixer. This extruder was mounted on a Geeetech A10T 3D printer via a WhamBam Mutant V2 mounting system, chosen for its quick-release feature, allowing fast switching of the mixer geometry. The Geeetech A10T controlled the extrusion process, while for the hotend, an external cartridge heater was used to heat up the hotend aluminum block. A controlling circuit was designed to manage the cartridge heater inside the mixer, the schematic used for connecting the Heater, REX Controller, Solid State Relay, and Thermocouple is shown in Fig. S2. The external heater was necessary since the selected cartridge heater operated on 110 V AC power. Figure 3 illustrates the experimental setup.

To assemble the mixer hot end with the desired mixing geometry for testing, the process begins by placing locating pins, the geometry, and screws into the top and bottom blocks. After assembling these blocks, the heater is inserted into the designated heating hole and secured with a set screw. Next, the nozzle is threaded into the bottom of the lower block. The final step involve wrapping the threads at both ends of the pipe nipple with

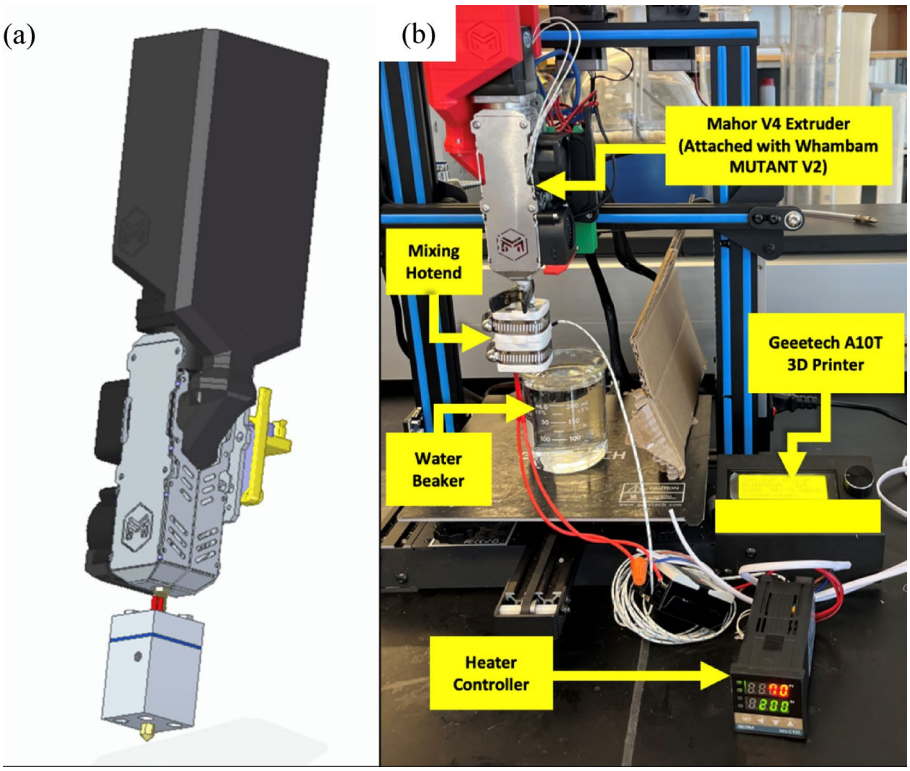


Fig. 3. Experimental setup, (**a**) the rendered setup in CAD, and (**b**) the actual assembly.

Teflon tape and threading the nipple into the input hole on the top block. Subsequently, the assembly is attached to the Mahor Extruder by threading the end of the pipe nipple into the threaded output hole at the bottom of the extruder. Finally, the entire assembly is mounted onto the 3D printer using the Whambam system. To assess the mixing quality of polymers in the extruded filament, cross-sections of the extrusions must be analyzed. Rapid cooling of the extrusion is essential to maintain the mixing state as it exits the nozzle. To achieve this, the nozzle is positioned just above a beaker of water, allowing for immediate cooling of the extruded material. This method not only helps preserve the cross-sectional mixing but also prevents the extrusion from necking down as it becomes longer and heavier since the cooled material is less pliable and more resistant to deformation. Furthermore, submerging the extrusion in water reduces the force pulling it from the nozzle due to buoyancy, contributing to a more consistent cross-sectional area throughout the process.

Machine learning analysis

Histogram analysis techniques

Histogram analysis is a fundamental image processing technique that examines the distribution of pixel intensities in an image, providing a comprehensive understanding of color distribution and mixing quality. In the context of evaluating the best mixer for combining red and blue liquids, histogram analysis involves quantifying the performance using several indices, including Entropy, Intersection Index, Chi-Square, and Bhattacharyya Index. Each of these measures plays a distinct role in assessing the uniformity and effectiveness of color mixing.

Entropy[24] measures the randomness or diversity of pixel intensities in the image histogram. High entropy suggests that the pixel values are highly diverse, with many intensity levels. On the other hand, low entropy suggests that certain intensities or colors dominate, resulting in less variety and potentially more predictability. Low entropy shows more uniformity and good mixing. By calculating the entropy of the histogram, we can evaluate how evenly the colors are blended, providing an objective metric to compare different mixers. The formula for entropy is given by:

$$H = -\sum p_i \log p_i \tag{1}$$

where p_i represents the probability of a pixel intensity i.

The Intersection Index[25] assesses the overlap between the histograms of the original colors (red and blue) and the resulting mixed image. A higher intersection value suggests significant retention of the original colors, while a lower value indicates better blending and creation of new mixed colors. This metric is particularly useful for visualizing how well the mixer eliminates the distinction between red and blue, achieving a seamless mixture. The intersection is computed as:

$$I = \sum min\left(H_R\left(i\right), H_B\left(i\right)\right) \tag{2}$$

where $H_R\left(i\right)$ and $H_B\left(i\right)$ are the histogram values for red and blue pixels, respectively.

The Chi-Square metric[26] compares the similarity between the expected and observed histograms. A lower Chi-Square value indicates greater similarity, suggesting that the observed histogram closely matches the expected distribution for an ideal mixture. This technique provides a statistical basis for quantifying the deviation of the observed mixing results from the desired uniform distribution. The formula for Chi-Square is:

$$\chi^2 = \sum \frac{\left(H_O\left(i\right) - H_E\left(i\right)\right)^2}{H_E\left(i\right)} \tag{3}$$

where $H_O\left(i\right)$ and $H_E\left(i\right)$ are the observed and expected histogram values, respectively.

The Bhattacharyya Index[27] provides insights into how closely the color distributions of the mixed image resemble the original colors. A higher Bhattacharyya coefficient suggests that the mixed image retains significant information about the original colors. The low Bhattacharyya index suggests the original color distribution of less present in the mixed image and it indicates effective mixing. This index is particularly effective in assessing the overall quality of mixing. The Bhattacharyya coefficient is calculated as:

$$B = \sum_i \sqrt{H_1\left(i\right).H_2\left(i\right)} \tag{4}$$

By combining these indices, the histogram analysis technique provides a multi-faceted evaluation of color mixing quality. Entropy captures the randomness and diversity of the mixture, Intersection Index measures the overlap of original and mixed colors, Chi-Square quantifies the deviation from an ideal mixture, and the Bhattacharyya Index evaluates the retaining of the original colors. These metrics offer a robust framework for identifying the most effective mixer for blending red and blue liquids.

Cluster analysis techniques

Cluster analysis is a powerful technique used to evaluate the quality of mixing by analyzing the distribution of color pixels in the image of a mixer. This approach involves grouping similar pixel colors into clusters and assessing how well the clusters represent the mixture. To identify the best-performing mixer, three commonly used clustering evaluation methods are applied: the Elbow Method, the Davies–Bouldin Index, and the Calinski–

Harabasz Index. These techniques provide insights into the optimal number of clusters and the effectiveness of mixing based on the distribution of color pixels.

The Elbow Method[28] is used to determine the optimal number of clusters by examining the total within-cluster sum of squares (WCSS), which measures the variance within each cluster. By plotting the WCSS against the number of clusters, the "elbow point" is identified as the point where the rate of decrease in WCSS slows down. This point represents the optimal number of clusters, indicating the most effective separation of mixed colors. In the context of mixer evaluation, a well-mixed image will exhibit a smooth transition in WCSS, reflecting consistent color blending.

The Davies–Bouldin Index[29] quantifies the compactness and separation of clusters, providing a measure of cluster quality. A lower Davies–Bouldin Index indicates better clustering, as it suggests compact clusters that are well-separated from each other. For mixer evaluation, this method helps determine how distinctly the red and blue colors blend to form new shades. The index is calculated as:

$$DB = \frac{1}{n}\sum_{i=1}^{n} \underset{j \neq i}{max} \left(\frac{\sigma_i + \sigma_j}{d_{ij}}\right) \tag{5}$$

where σ_i and σ_j represent the average distances of points within clusters i and j to their centroids, and d_{ij} is the distance between the centroids of clusters i and.

The Calinski–Harabasz Index[30], also known as the Variance Ratio Criterion, evaluates the ratio of between-cluster variance to within-cluster variance. A higher index value indicates better-defined and more compact clusters. For mixer evaluation, this method assesses how effectively the mixer reduces the distinctness of the original red and blue colors while creating new, uniform color clusters. The index is calculated as:

$$CH = \frac{trace\,(B_k)}{trace\,(W_k)} \frac{(n-k)}{(k-1)} \tag{6}$$

where B_k is the between-cluster scatter matrix, W_k is the within-cluster scatter matrix, n is the number of data points, and k is the number of clusters.

By combining these cluster analysis techniques, the evaluation of mixers becomes more precise and comprehensive. The Elbow Method identifies the optimal number of clusters, the Davies–Bouldin Index measures the compactness and separation of clusters, and the Calinski–Harabasz Index assesses the variance distribution. Together, these methods provide a robust framework to analyze color pixel clustering and determine the best mixer for achieving uniform and effective blending of red and blue liquids.

Standard index analysis techniques

Standard index analysis techniques provide a quantitative framework for assessing the effectiveness of mixers by evaluating the quality and uniformity of the mixed color distribution in images. These techniques involve calculating indices that measure pixel similarity, color consistency, and structural coherence. The following indices are utilized in this analysis: Structural Similarity Index (SSIM), Mutual Information Index (MI), Mean Square Error (MSE), Normalized Cross-Correlation (NCC), and Standard Deviation Uniformity (SDU).

The Structural Similarity Index (SSIM)[31] measures the perceived quality of an image compared to a reference image. It evaluates structural similarity based on luminance, contrast, and structure. In the context of mixer evaluation, SSIM assesses how closely the mixed image resembles the expected uniformly mixed result. It is defined as:

$$SSIM\,(x,y) = \frac{(2\mu_x\mu_y + C_1)\,(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)\,(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{7}$$

where μ_x and μ_y are mean intensities, σ_x^2 and σ_y^2 are variances σ_{xy} is the covariance and C_1, C_2 are constants. A higher SSIM value indicates a more uniform mixture.

The mutual information index (MI)[32] measures the amount of shared information between two images, reflecting the degree of dependence between their pixel distributions. For mixer evaluation, MI quantifies how well the red and blue colors have blended to form a new distribution. Higher mutual information indicates that the mixed image retains significant details on the original colors and the absence of new colors or mixed colors. So, a lower value of MI indicates a good mixer. MI is calculated as:

$$MI\,(x,y) = \sum_{i,j} p\,(x_i, y_i)\, log \frac{p\,(x_i, y_i)}{p\,(x_i)\,p\,(y_i)} \tag{8}$$

where $p\,(x_i, y_i)$ is the joint probability distribution of x and y and $p\,(x_i)\,, p\,(y_i)$ are their marginal probabilities.

The mean square error (MSE)[33] evaluates the average squared difference between pixel intensities of the mixed image and an ideal reference image. It serves as a measure of deviation, with a lower MSE indicating better mixing and uniformity. MSE is calculated as:

$$MSE = \frac{1}{N}\sum_{i=1}^{N} (x_i - y_i)^2 \tag{9}$$

where x_i and y_i are the pixel intensities of the mixed and reference images, respectively, and N is the total number of pixels.

The normalized cross-correlation (NCC)[34] evaluates the similarity between two images by comparing their pixel intensity patterns. For mixer evaluation, it measures the degree of alignment between the red and blue pixel distributions after mixing. The NCC is calculated as:

$$NCC\left(x,y\right)=\frac{\sum_{i=1}^{N}\left(x_i-\overline{\overline{x}}\right)\left(y_i-\overline{\overline{y}}\right)}{\sqrt{\sum_{i=1}^{N}\left(x_i-\overline{\overline{x}}\right)^2\sum_{i=1}^{N}\left(y_i-\overline{\overline{y}}\right)^2}} \tag{10}$$

where $\overline{\overline{x}}$ and $\overline{\overline{y}}$ are the mean pixel intensities of x and y, respectively. A higher NCC value indicates a more uniform and effective mixture.

The standard deviation uniformity (SDU)[35] evaluates the uniformity of pixel intensities within the mixed image by analyzing the standard deviation of color intensity. A lower standard deviation indicates a more consistent and homogeneous mixture. It is calculated as:

$$SDU=\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(x_i-\overline{\overline{x}}\right)^2} \tag{11}$$

where x_i represents pixel intensities, $\overline{\overline{x}}$ is the mean intensity, and N is the total number of pixels. SDU provides insights into the dispersion of colors in the mixer image.

These measuring index techniques collectively provide a comprehensive evaluation of mixer performance. SSIM and MI focus on structural similarity and information integration, MSE and NCC quantify deviations and alignment, while SDU emphasizes uniformity. By analyzing these indices, the best-performing mixer can be identified based on its ability to achieve a uniform and effective blend of red and blue colors. The steps followed in the machine learning analysis are illustrated in Fig. S3.

Experimental data collection

Once the extruder and mixer are heated, the extruder is operated until the mixer is filled with material. The manual control on the 3D printer is accessed, and a setting is configured to extrude 1000 mm. The output nozzle is monitored for extrusion, with approximately 3000 mm typically needed for a steady flow. Initial extrusion may show inconsistencies; however, once stability is achieved, a sample about 150 mm long is extruded into the water, broken off, and labeled. The procedure is done five times for each intermixer design. For a control test, the extruder is run normally with the same pellets and temperature, utilizing a nozzle directly in the output hole instead of the mixer hot end. This approach is faster, as there is no empty volume to fill.

Image data quality and processing

During the microscope image analysis, some challenges arose, including inconsistent imaging scale due to variations in lens distance, deformations from the cutting process, and slight differences in specimen diameter, which ranged from 0.75 to 1.00 mm. The microscope images are shown in Fig. 4. The analysis revealed differences in the performance of various mixer designs. The Split Path Mixer and Helix Array demonstrated good blending, with homogeneous color distributions observed throughout the cross-sections. On the other hand, the Full Turn Helix, Half Moon, and Cross Bars designs showed slightly less good blending. In the result section, the performance of these mixers is demonstrated.

During the process of capturing the images, an external yellow light source was applied, which introduced a noticeable yellowish tint to the photos and this additional color changed the actual color of the mixer. This unintended color cast can interfere with the accurate analysis of the red and blue color distributions in the images. To address this issue, it is necessary to apply image processing techniques to remove the excess yellow light and restore the original color balance. By isolating the yellow component in the image, typically represented as a combination of red and green channels in RGB color space, these techniques can accurately subtract or neutralize the added yellow light as appears in Fig. 5. This correction ensures that the images more accurately reflect the actual colors of the specimens, enabling more precise analysis and interpretation of the mixing and blending performance.

Intermixer performance evaluation by machine learning

This section presents the results of the image analysis conducted to evaluate the performance of various mixers in blending two distinct colors of PLA. The analysis aimed to quantitatively assess the mixing quality by examining images of the mixed PLA using three complementary approaches: histogram analysis techniques as in Table 1, cluster analysis techniques as in Table 2, and standard index analysis techniques as in Table 3. Each approach offers unique insights into the distribution, clustering, and similarity of the color components within the images, providing a comprehensive evaluation of the mixer performance.

The outcomes represented in Table 1 provide insights into the performance of different types of mixers in terms of their ability to mix red and blue PLA, as measured by histogram-based metrics: Entropy, Intersection, Chi-square, and Bhattacharyya Index. Each of these metrics provides a unique perspective on the mixing effectiveness.

The Half Moon, Full Turn Helix and Split Mixers consistently perform better across all metrics, suggesting they are the most effective mixer shapes in achieving a uniform and well-mixed PLA. The Control type shows

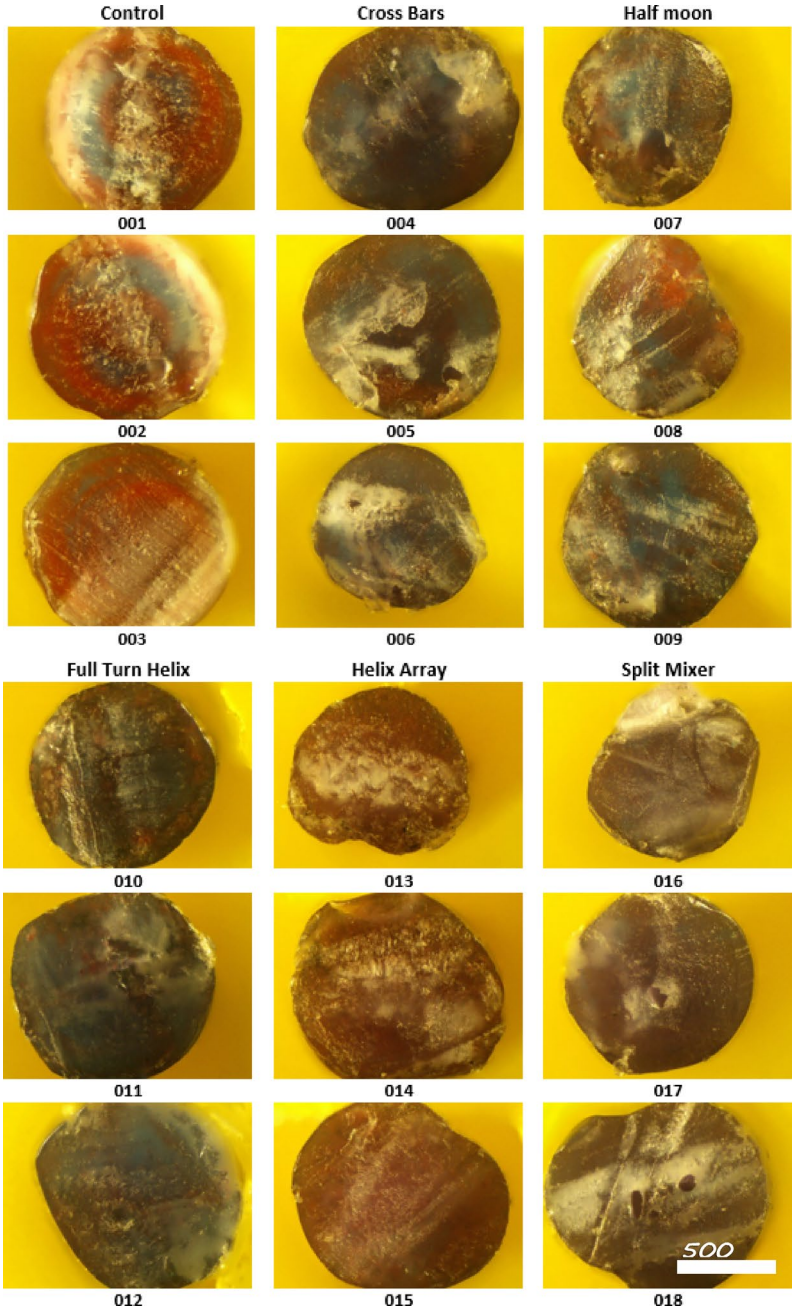


Fig. 4. The results of the microscope inspection of the six mixer samples [All images follow the same scale].

the weakest performance, emphasizing the necessity of a structured mixing mechanism for effective blending. Metrics like Entropy and Intersection highlight the uniformity and overlap, while Chi-square and Bhattacharyya Index provide insights into deviations and similarity, offering a comprehensive understanding of the mixers' performance. This analysis indicates that advanced mixer designs significantly enhance the mixing efficiency compared to the Control setup.

Table 2 summarizes the clustering performance of various mixer types based on three evaluation methods: K-Mean, Davies–Bouldin Method (DBM), and Calinski–Harabasz Method (CHM). These methods provide insights into the optimal number of clusters detected in the mixing process of red and blue liquids as appears in Fig. 6.

Overall, the results show that while the Control mixer produces simpler and more uniform outcomes, advanced mixers like Helix Array, Split Mixer, and Full Turn Helix demonstrate superior performance with more complex and effective mixing patterns, making them better suited for achieving sophisticated mixing outcomes.

The outcomes presented in Table 3 illustrate the performance of various mixers in blending red and blue liquids, evaluated using multiple analytical indices: Structural Similarity Index (SSIM), Mutual Information Index (MI), Mean Square Error (MSE), Normalized Cross-Correlation (NCC), and Standard Deviation

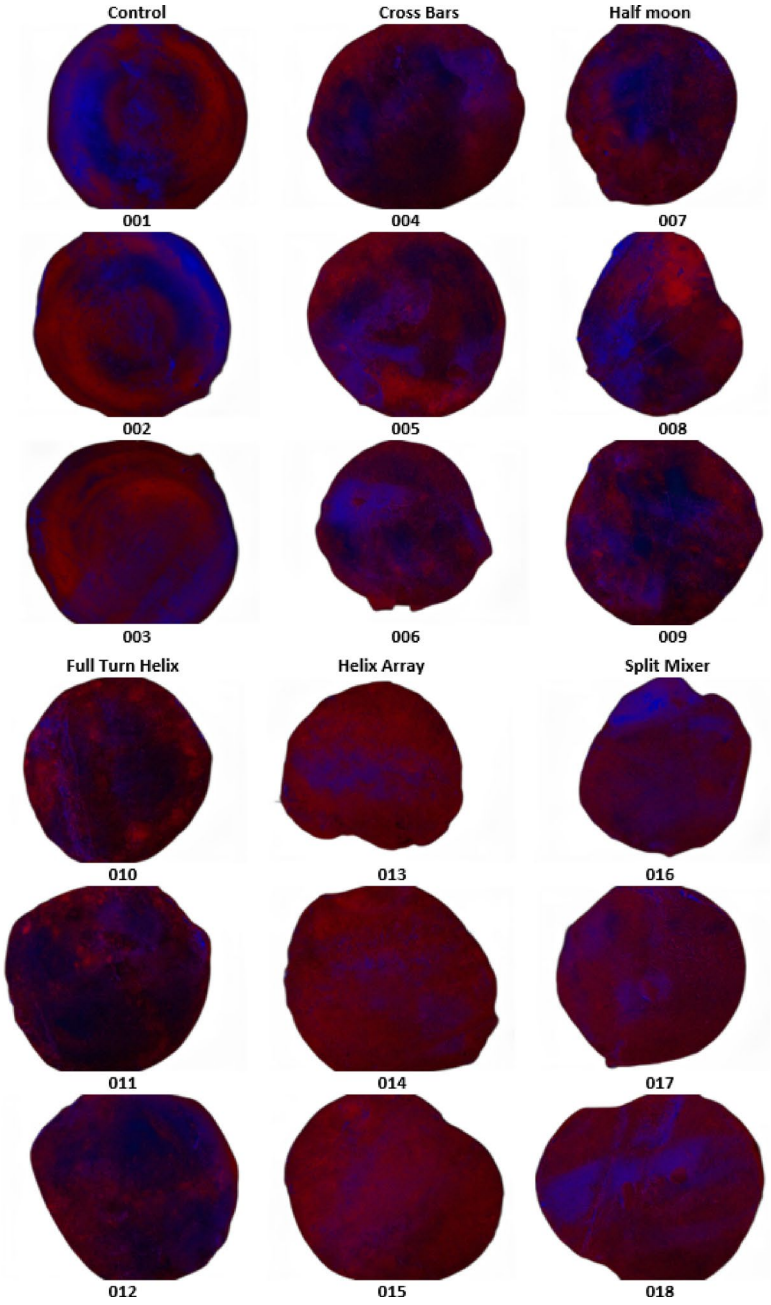


Fig. 5. The results of applying image processing techniques to restore the original color balance.

Uniformity (SDU). These indices reveal significant differences in mixing effectiveness between the control type and specialized mixers, with each index providing distinct insights into the quality of mixing. High values in SSIM, and NCC generally indicate better mixing performance, while lower MSE, MI, SDU values reflect reduced errors, signifying more uniform and effective mixing.

Histograms represent the distribution of pixel intensities and analyzing them provides valuable insight into how the colors are distributed across the image. For each image, the histogram for a color (such as red or blue) represents the frequency of each pixel intensity for that color channel, ranging from 0 (no intensity) to 255 (maximum intensity). The histogram illustration of mixer performance is shown in Fig. 7.

Each mixer type has 3 histogram images, and each image contains data on the red and blue color intensities. The purpose of the histograms in the context of these mixers is that they represent the distribution of the intensity of red and blue colors in the images, reflecting how well the mixer is blending these colors. The skewness of the histogram refers to whether the distribution leans toward the high or low end of the intensity scale. Look for peaks (dominant red intensity), spread (variation), and skewness (distribution shape). Narrow, high peaks indicate dominance, wide peaks indicate diversity. Same analysis applies to the blue histogram. Compare how blue is mixed with red. Assess balance, dominance, and the mixing patterns between the two colors. Describe

Sample	Mixer type	Entropy	Intersection	Chi-square	Bhattacharyya
001	Control	3.77	2.23	275.66	0.27
002	Control	3.71	2.24	141.68	0.25
003	Control	3.67	1.95	16.95	0.4
004	Cross Bars	3.74	2.82	0.76	0.15
005	Cross Bars	3.62	2.14	2.36	0.24
006	Cross Bars	2.89	1.7	0.67	0.16
007	Halfmoon	3.06	1.86	0.35	0.13
008	Halfmoon	3.04	1.76	0.42	0.13
009	Halfmoon	3.49	2.41	0.2	0.1
010	Full Turn Helix	3.29	1.79	2.37	0.29
011	Full Turn Helix	3.55	2.6	0.51	0.17
012	Full Turn Helix	3.54	2.71	0.28	0.13
013	Helix Array	2.87	1.21	23.58	0.49
014	Helix Array	3.51	1.23	73.15	0.66
015	Helix Array	3.54	1.24	66.26	0.66
016	Split Mixer	2.91	1.59	6.6	0.25
017	Split Mixer	3.26	1.58	8.2	0.38
018	Split Mixer	3.9	2.3	9.22	0.36

Table 1. The results of histogram analysis.

Sample	Mixer type	K-mean	DBM	CHM
001	Control	3	3	10
002	Control	3	3	10
003	Control	3	2	5
004	Cross Bars	3	2	7
005	Cross Bars	3	2	4
006	Cross Bars	3	2	4
007	Half moon	3	2	3
008	Half moon	3	2	7
009	Half moon	3	2	5
010	Full Turn Helix	3	2	2
011	Full Turn Helix	3	2	2
012	Full Turn Helix	3	2	3
013	Helix Array	2	2	3
014	Helix Array	2	2	2
015	Helix Array	2	2	2
016	Split Mixer	3	2	4
017	Split Mixer	2	2	2
018	Split Mixer	3	2	3

Table 2. The results of clustering analysis.

how each mixer influences the red and blue histograms, helping to identify which mixers produce balanced or unbalanced color mixing.

By conducting a detailed analysis of the histograms, one can draw conclusions about the effectiveness and characteristics of each mixer, and how they impact the distribution of red and blue colors in the images.

Figures 8a–f and 9a–f provide a detailed comparative analysis of color mixing performance between the Control mixer (No mixer) and the Cross Bars mixer, demonstrating their effectiveness in blending red and blue materials. Each subfigure presents a different aspect of the mixing process, including color distribution, density-intensity relationships, and quantitative assessments of newly formed colors.

Figures 8a and 9a illustrate the segmentation of colors formed after the mixing of red and blue materials. The horizontal axis represents blue intensity, and the vertical axis represents red intensity, while the colors are classified into three clusters, each representing a different degree of mixing. The RGB color codes assigned to each cluster indicate the specific shades formed due to blending. The relative positioning and spread of these clusters indicate how effectively the colors have merged, with well-mixed samples exhibiting smoother transitions and less distinct separation between clusters. Additionally, the RGB color codes associated with each cluster

Sample	Mixer type	SSIM	MI	MSE	NCC	SDU
001	Control	0.53	0.9	72.27	0.12	35.61
002	Control	0.54	0.94	70.94	0.07	35.04
003	Control	0.51	0.82	76.16	0.26	28.16
004	Cross Bars	0.54	0.65	70.9	0.48	25.92
005	Cross Bars	0.55	0.73	67.62	0.47	31.57
006	Cross Bars	0.68	0.73	49.04	0.66	30.48
007	Half moon	0.63	0.75	55.34	0.5	27.43
008	Half moon	0.65	0.79	51.9	0.43	34.8
009	Half moon	0.56	0.66	64.61	0.52	27.77
010	Full Turn Helix	0.56	0.67	62.03	0.51	25.29
011	Full Turn Helix	0.55	0.61	64.79	0.55	22.44
012	Full Turn Helix	0.56	0.66	67	0.52	23.82
013	Helix Array	0.58	0.71	58.93	0.52	29.92
014	Helix Array	0.42	0.61	81.75	0.47	26.09
015	Helix Array	0.44	0.66	81.42	0.63	27.63
016	Split Mixer	0.67	0.77	50.62	0.65	31.85
017	Split Mixer	0.58	0.73	64.81	0.71	29.76
018	Split Mixer	0.48	0.63	80.8	0.48	31.88

Table 3. The results of standard index analysis.

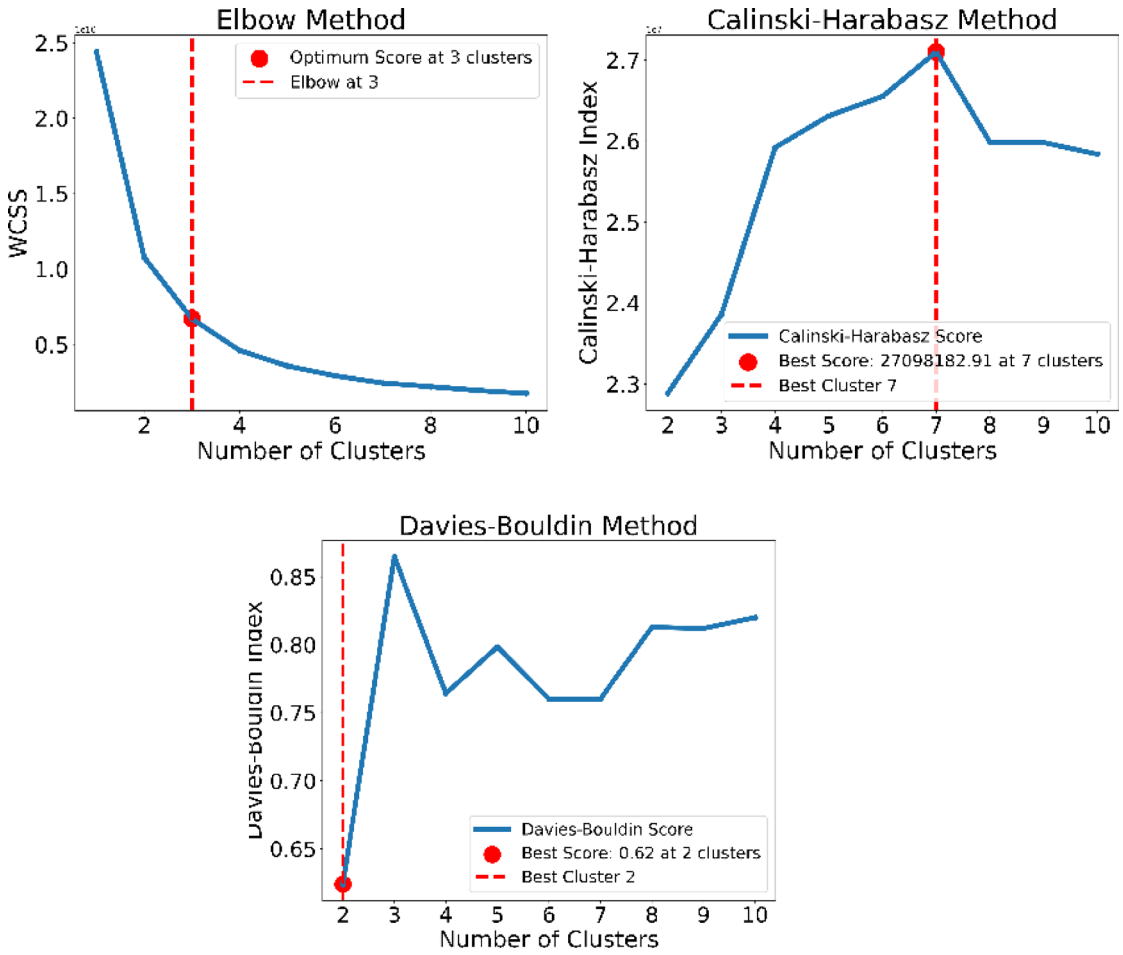


Fig. 6. Clustering performance evaluation methods.

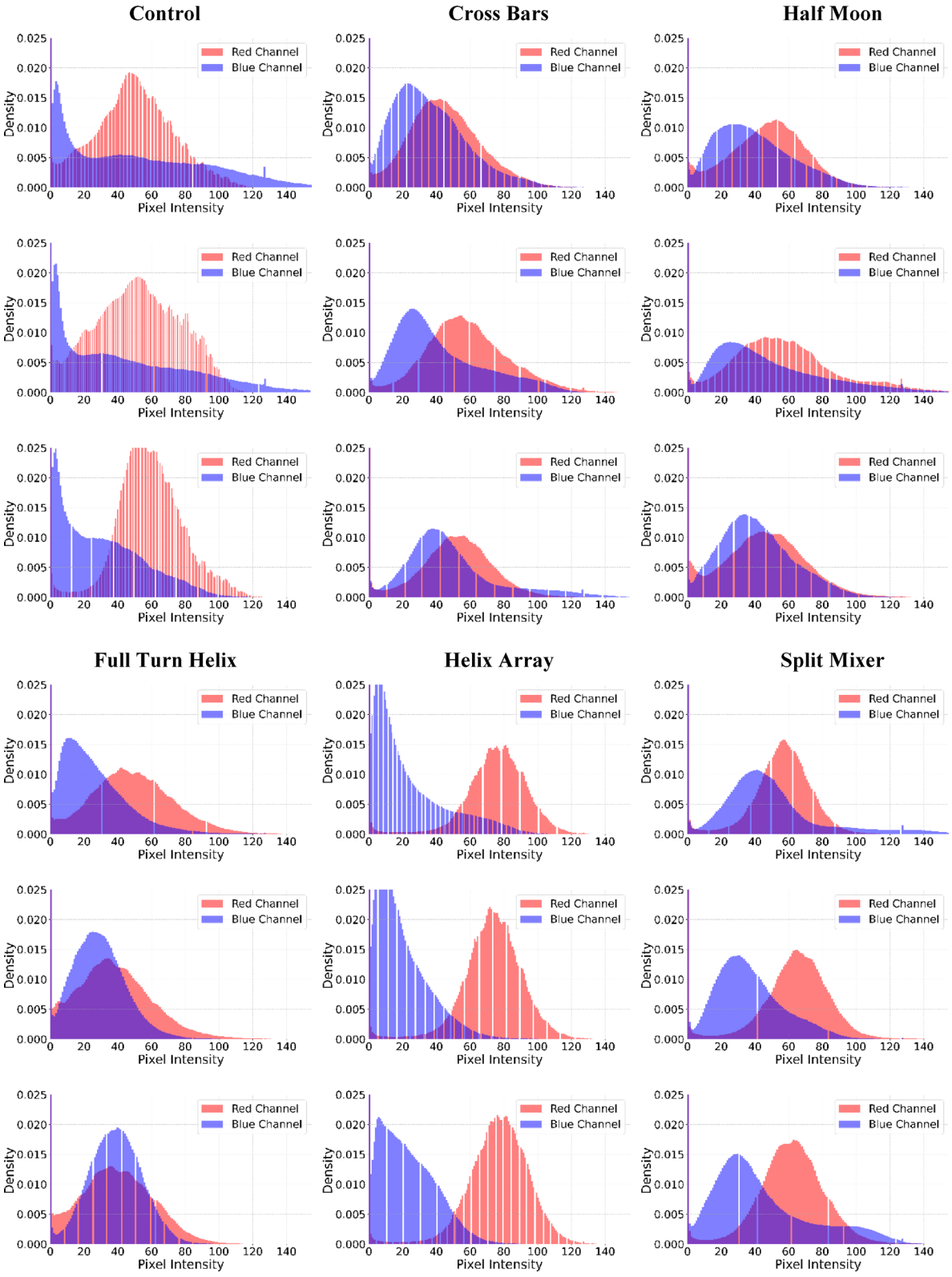


Fig. 7. Histogram Illustration of different mixers performance.

provide valuable insights into the extent of color blending. These codes indicate the specific shades formed due to the interaction between red and blue intensities, revealing whether the mixing process has resulted in a homogeneous color blend or retained distinct patches of the original colors.

Figures 8b–d and 9b–d provide density-intensity plots, which further examine how red and blue colors interact in different mixing scenarios. Here, the x-axis represents the intensity values of red and blue colors,

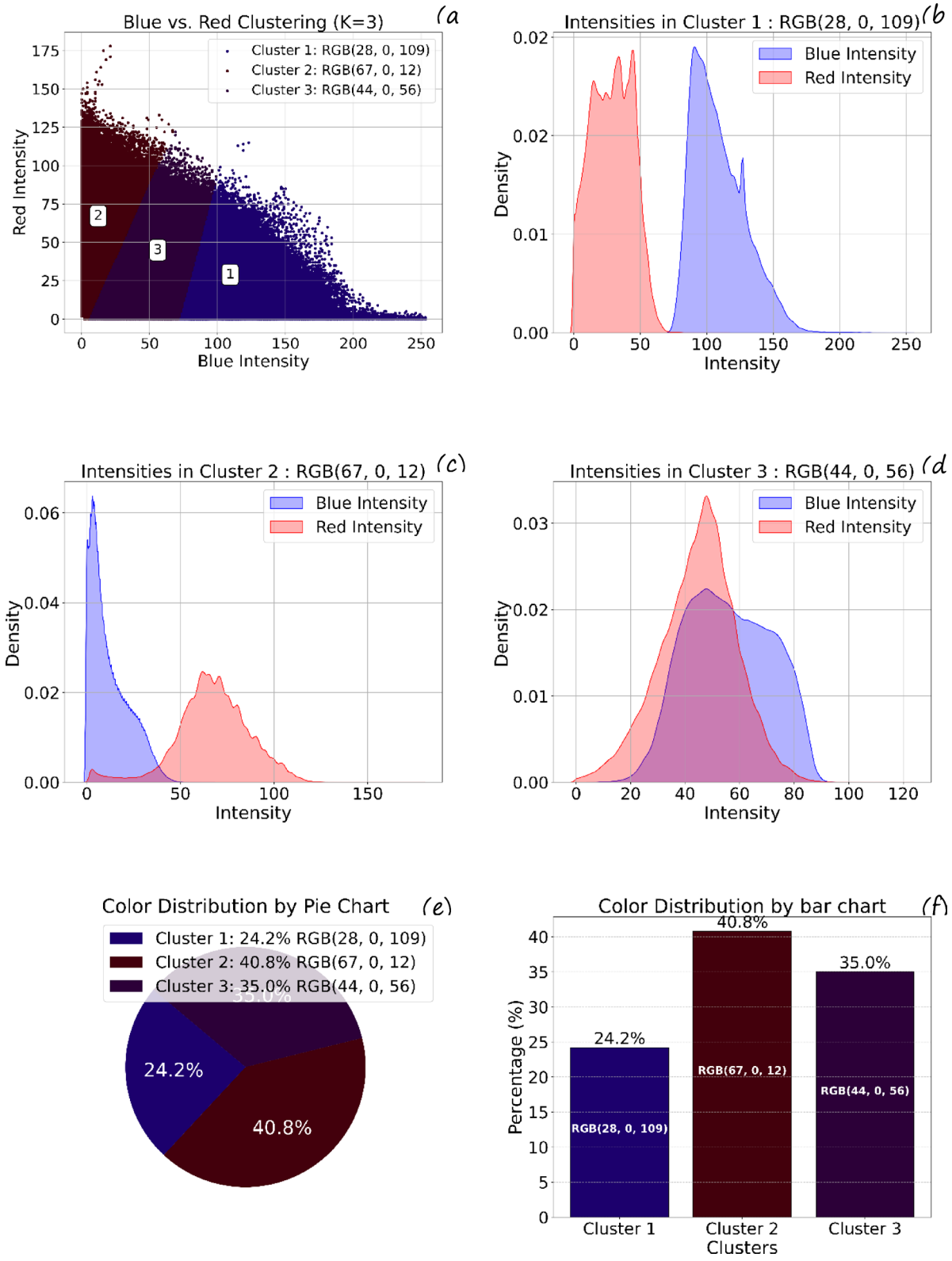


Fig. 8. The General Mixing color distribution without specific mixer, (**a**) shows the red and blue intensities of each of the three clusters; (**b**) the RGB intensities in case of Cluster 1; (**c**) the RGB intensities in case of Cluster 2; (**d**) the RGB intensities in case of Cluster 3; (**e**) shows the three clusters as a pie chart; and (**f**) the bar chart of the color distribution for each cluster.

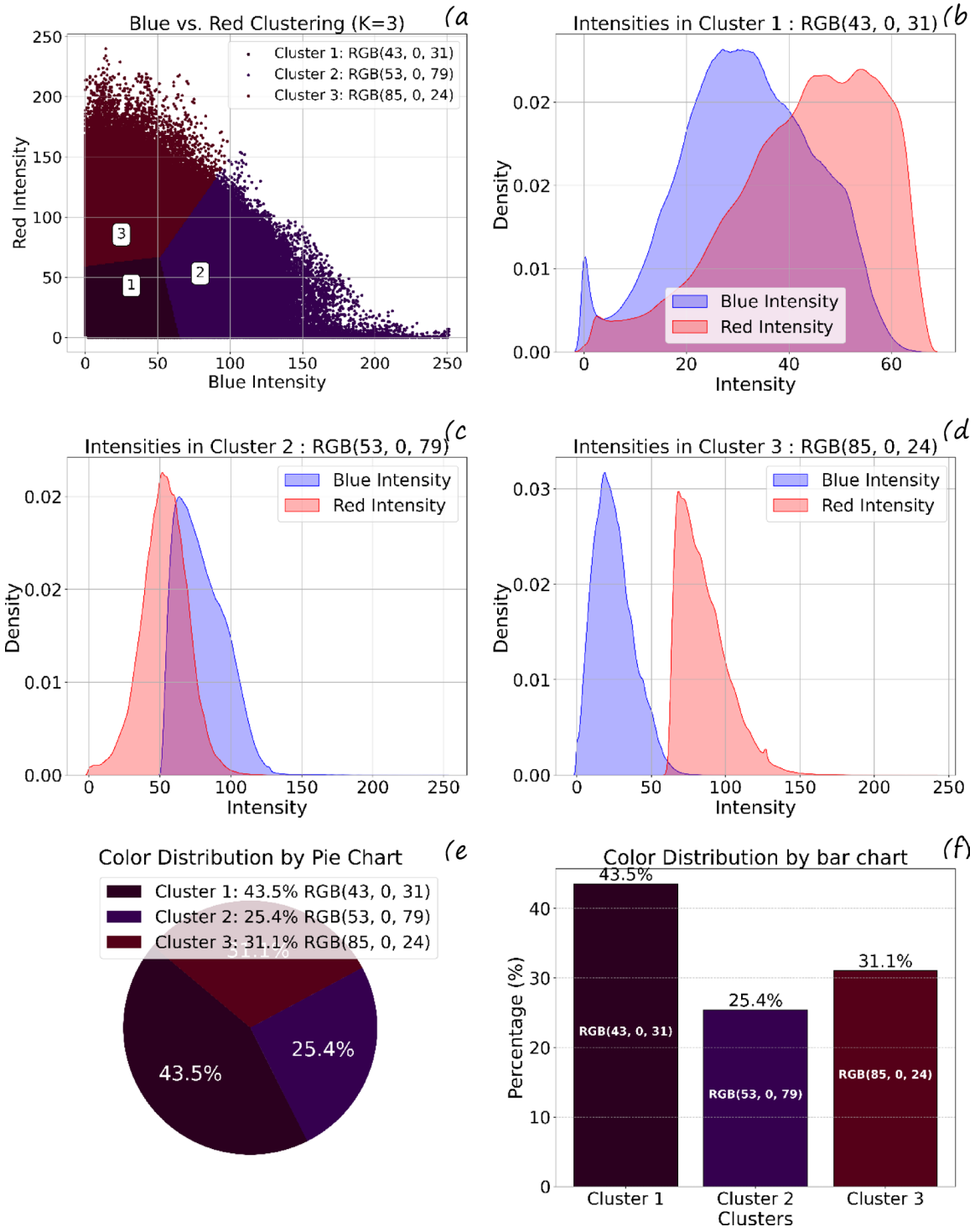


Fig. 9. The Mixing Color Distribution using Mixers such as Cross Bars, (**a**) shows the red and blue intensities of each of the three clusters taking into consideration the Cross Bars mixer; (**b**) the RGB intensities in case of Cluster 1; (**c**) the RGB intensities in case of Cluster 2; (**d**) the RGB intensities in case of Cluster 3; (**e**) shows the three clusters as a pie chart; and (**f**) the bar chart of the color distribution for each cluster.

and the y-axis represents the density of red and blue pixels in the sample images. Each subfigure breaks down a specific color cluster from Figs. 8a and 9a, revealing how red and blue overlap and mix. In Fig. 8b–d (Control Mixer—No Mixer), there is less overlap between the red and blue densities, signifying poor blending and distinct color separation. In Fig. 9b–d (Cross Bars Mixer), the density plots show greater overlap, indicating better integration of the two colors and improved mixing efficiency. The greater the overlap between red and blue

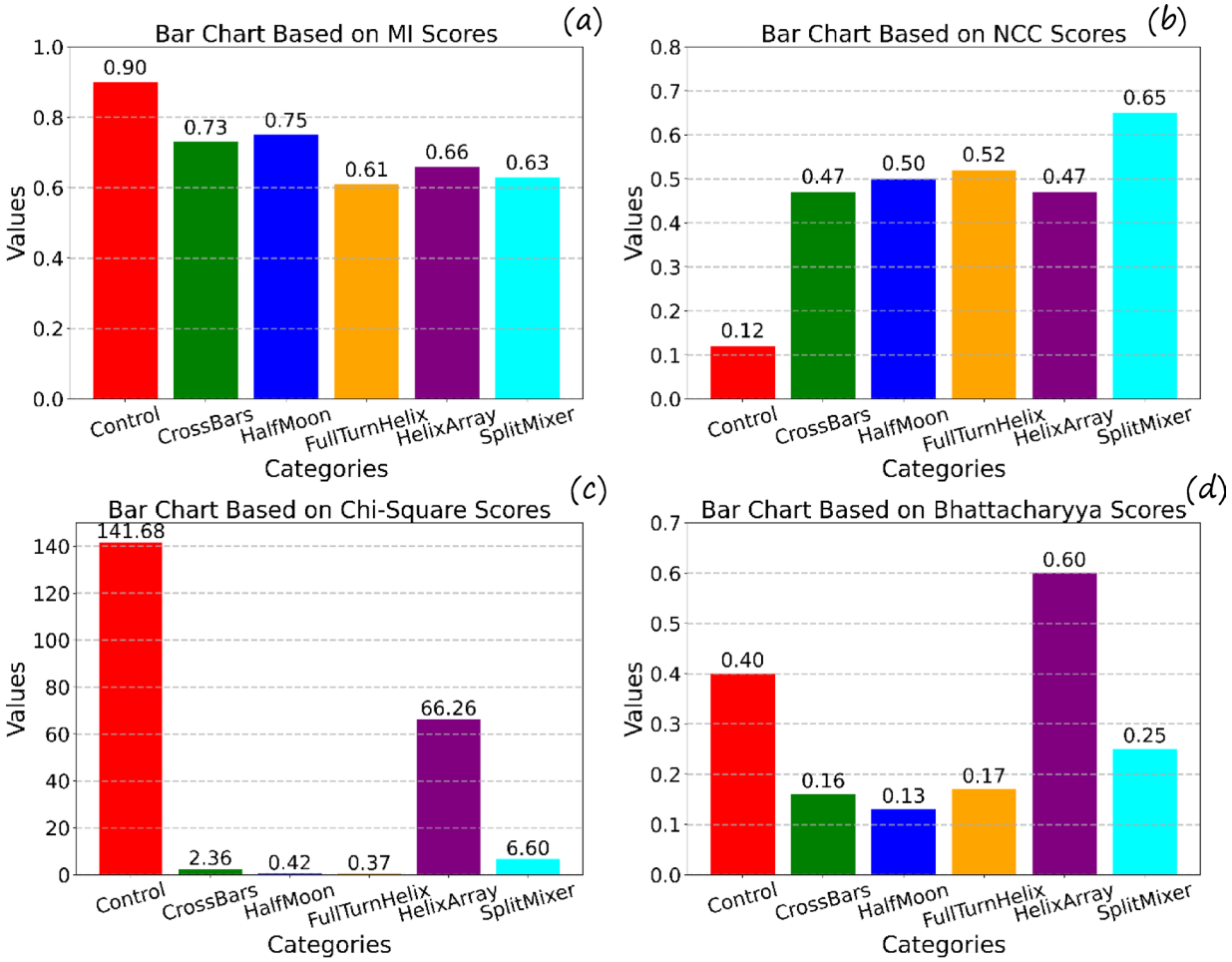


Fig. 10. The Overall performance of The Mixers in the Bar Chart, (**a**) based on MI scores; (**b**) based on NCC scores; (**c**) based on Chi-Square scores; (**d**) based on Bhattacharyya scores.

densities, the more uniform the color distribution, proving that the Cross Bars mixer facilitates superior material dispersion compared to the Control case.

Figures 8e and 9e present pie charts representing the proportional distribution of newly created colors after mixing. Similarly, Figs. 8f and 9f display bar charts that further quantify the percentage and pixel count of newly created colors in the sample images. This visualization helps in identifying the relative percentages of different color shades formed due to the blending of red and blue. The pie and bar chart segments represent the proportion of each color present in the final mixture, illustrating how efficiently the mixer distributes and blends the two base colors.

The complete analysis of Figs. 8a–f and 9a–f clearly shows that the Cross Bars mixer performs significantly better than the Control mixer (No Mixer)**.** When red and blue materials are mixed, they combine in varying proportions, resulting in the formation of new intermediate colors. The intensity and distribution of these colors depend on the degree of blending achieved during the mixing process. As a result, the final output exhibits a gradient of hues, ranging from deep maroon to lighter shades of violet, depending on the local concentration of red and blue components. Furthermore, it is important to consider the influence of color filtering on the visual appearance of the samples. Specifically, when yellow light is removed from the sample images, a noticeable reduction in overall color intensity occurs, with a significant loss of red saturation. This alteration affects the perceived colors in the blended material, making them appear darker and more muted. These results highlight the importance of using structured mixing elements like the Cross Bars design to enhance color uniformity, material homogeneity, and overall mixing efficiency in multi-material extrusion applications.

To assess the effectiveness of different mixer designs, in Fig. 10, four key statistical and machine learning-based metrics were analyzed: Mutual Information (MI), Normalized Cross-Correlation (NCC), Chi-Square, and Bhattacharyya Scores. Each of these metrics provides unique insights into the degree of uniformity in the mixed material. The six tested mixer designs, Control (no mixer), Cross Bars, Half Moon, Full Turn Helix, Helix Array, and Split Path Mixer, were evaluated using these four methods. In Fig. 10a, the MI values reflect the amount of retained correlation in the extruded material, where lower MI values indicate better mixing. The results showed that the Split Path Mixer and Helix Array achieved the lowest MI values, confirming their superior ability to break material clusters and enhance uniform blending. In contrast, the Cross Bars and Half Moon mixers

exhibited higher MI values, indicating less effective mixing. Most importantly, all mixers performed better than the Control case (no mixer), which had the highest MI value, demonstrating that structured mixing elements improve material dispersion. In Fig. 10b, NCC measures the spatial consistency of material distribution, where higher values indicate better mixing. The Split Path Mixer achieved the highest NCC value, signifying excellent polymer dispersion and uniform color blending. Meanwhile, the Cross Bars, Half Moon, Full Turn Helix, and Helix Array mixers showed slightly lower NCC values, indicating relatively weaker mixing performance. However, all mixer designs outperformed the Control case, which recorded the lowest NCC value, confirming that mixing mechanisms significantly improve material homogeneity. In Fig. 10c, the Chi-Square metric assesses variations in color distribution, where lower values indicate better mixing. The Full Turn Helix and Half Moon mixers exhibited the lowest Chi-Square values, confirming their effectiveness in achieving uniform blending. Conversely, the Split Path Mixer and Cross Bars showed slightly higher Chi-Square values, indicating a moderate degree of phase separation. Notably, the Helix Array demonstrated the highest Chi-Square value, suggesting significant variation and poor mixing performance. Despite these variations, all mixers performed better than the Control case, which had the highest Chi-Square value, reinforcing the role of structured mixing elements in improving extrusion quality. In Fig. 10d, the Bhattacharyya score measures the similarity between color distributions, where lower scores indicate better mixing. The Full Turn Helix, Cross Bars, Split Path Mixer, and Half Moon mixers showed the lowest Bhattacharyya scores, confirming their ability to achieve highly uniform mixing. In contrast, the Helix Array exhibited the highest Bhattacharyya score, even exceeding the Control case, indicating that this design resulted in poor material dispersion and severe phase separation. However, four mixers outperformed the Control case, demonstrating the benefits of incorporating structured mixing mechanisms

.

The comprehensive analysis of all four statistical and machine learning-based metrics highlights the effectiveness of structured mixer designs in improving material homogeneity in multi-material extrusion. While the Full Turn Helix, Split Path Mixer, and Half Moon mixers consistently performed well across multiple metrics, the Helix Array exhibited poor mixing performance, sometimes even worse than the Control case. Most importantly, all tested mixers significantly improved mixing compared to the Control case (extrusion without a mixer), confirming that structured mixing mechanisms enhance polymer dispersion, minimize phase separation, and improve overall extrusion quality.

RGB color blending limitations for thermomechanical mixing

RGB-based color blending is widely used in digital imaging and visualization due to its simplicity and visual naturality. However, as an indicator of the quality of polymer mixing, it suffers from certain constraints that weaken its potential as an alternative to thorough thermomechanical blending. While it provides a visual approximation of uniformity of colors, RGB blending primarily measures pigment dispersion rather than a direct view of the resulting molecular blending, viscosity profile, or mechanical homogeneity of polymer melts[36].

One of the primary constraints is the decay of component data: as pigments or materials are mixed in larger numbers, visualization of distinct constituents becomes increasingly problematic, especially more than two or three blended colors. RGB blending suffers from perceptual nonlinearity, in that blended colors do not readily correspond to human perception of blended coloration, and misinterpretation error can occur. Furthermore, the method becomes less practical to interpret with nuance, when many variables are encoded to a single composite color, information-carrying capacity is lost, and discrimination of each feature becomes harder to distinguish. Even using broad-apart colors or opponent colors in the RGB spectrum fails to drastically enhance interpretability[36,37].

These limitations make RGB blending an inadequate substitute for direct thermomechanical measurements, such as rheological testing, spectroscopy, or thermal imaging, which provide more substantial data on shear history, temperature homogeneity, and gradients in properties.

Conclusion

This study successfully evaluated and compared the performance of different mixer designs integrated into a coaxial extrusion system, aiming to enhance the blending efficiency of multi-colored PLA (polylactic acid) pellet mixtures. By employing a combination of microscopic imaging, image processing, and machine learning-based analysis, the study identified key factors influencing mixing uniformity and assessed the effectiveness of each mixer configuration.

Among the tested designs, the Split Path Mixer and Full Turn Helix demonstrated superior blending performance, achieving a highly uniform color distribution across the cross-sections of extruded samples. These mixers effectively reduced color segregation, ensuring a well-integrated polymer blend. In contrast, the Half Moon, Helix Array, and Cross Bars mixers exhibited incomplete mixing, with visible color separation and non-uniform dispersion, leading to their exclusion from further optimization due to their inconsistent performance.

To quantify mixing efficiency, machine learning and statistical analysis techniques were applied, including Histogram analysis, Clustering techniques, and Standard index methods, which confirmed significant improvements in color uniformity in the Split Path Mixer and Full Turn Helix compared to other designs. All five mixers consistently outperformed the Control mixer (No mixer), proving that structured mixing elements play a vital role in achieving homogeneous extrusion outputs.

These findings underscore the critical importance of advanced mixing mechanisms in multi-material additive manufacturing and functional polymer processing. Future research could focus on examining the blending of different materials which will focus more on thermomechanical analysis, and further optimizing extrusion parameters, such as temperature, feed rate, shear forces, to enhance mixing performance, refining mixer geometries to maximize material dispersion efficiency while maintaining scalability for industrial

applications, exploring AI-driven optimization methods to automate mixer design improvements based on real-time experimental feedback. By continuing to refine these mixer designs, multi-material extrusion technology can be significantly advanced, leading to higher-quality prints, improved mechanical properties, and greater versatility in additive manufacturing applications.

Data availability

The datasets generated and analyzed during the current study are available in the Mendeley Data repository, under the following https://doi.org/10.17632/664t5kcn7p.1. This includes raw microscope images, machine learning analysis scripts, and supplementary materials.

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Author contributions

A.R., C.S., D.H., and E.D. contributed equally. Conceptualization, A.R., C.S., D.H., and E.D.; Methodology, A.R., C.S., D.H., and E.D.; Validation, A.R., C.S., D.H., and E.D.; Formal Analysis, R.E., and M.H.; Investigation, R.E., and M.H.; Resources, R.E.; Data Curation, R.E.; Writing—Original Draft Preparation, R.E., and M.H.; Writing—Review and Editing, R.E., M.H., and M.A.H.K.; Visualization, M.H.; Supervision, M.A.H.K.; Project Administration, M.A.H.K.; Funding Acquisition, M.A.H.K.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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